Substitution is compatible with parallel beta-eta reduction in both arguments

The Open Logic Project contributors

openlogic:lambda-calculus/church-rosser:local-snapshot-20261013;
CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the delegation of the first four cases names its source lemma inside the same unit, and the new eta case is fully expanded; nothing is left implicit beyond the cited parallel argument.

Editorial note. The label lem:comp names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Editorial note. OLP reuses this label inside the unit; the card is identified by its statement span (source lines 712-715) and the claim key carries the statement start line.

Allowed standard prerequisites (editorial).

- Inductive definition of parallel beta-eta reduction. Same unit, \ollabel{defn:beredpar} with rules \ollabel{defn:beredpar1}-\ollabel{defn:beredpar5} (section parallel-beta-eta-reduction). Five-rule derivation induction; the new eta rule is the fifth.
- The beta-case bookkeeping of the parallel substitution lemma. Same unit, \ollabel{lem:comp} of the parallel-beta-reduction section (source line 825). The author transfers the first four cases verbatim to the beta-eta relation with 'exactly like those in' that lemma; only the fifth rule needs new work.

Source reference. Original definition defn-beredpar: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 669–669. Prerequisite pointer invoked at proof line(s) 726; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-beredpar5: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 680–680. Prerequisite pointer invoked at proof line(s) 721; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If $M \to_\beta M'$ and $R \to_\beta R'$, then $M[R/y] \to_\beta M'[R'/y]$.*

Proof. By induction on the !!derivation of $M \to_\beta M'$.

The first four cases are exactly like those in 0.1. If the last rule is defn-beredpar5, then M is $\lambda[x][Nx]$, M' is N' for some x and N' where $x \notin FV(N)$, and $N \to_\beta N'$. We want to show that $(\lambda[x][Nx])[R/y] \to_\beta N'[R'/y]$, i.e., $\lambda[x][N[R/y]x] \to_\beta N'[R'/y]$. It follows by defn-beredpardefn-beredpar5 and the induction hypothesis. □